PROVAL co.,Ltd

Calculation Sheet

Calculation header

Identifier	*New calculation (1)*

Medium selection and state

Medium	*251127-02 Test 01*
State	*Liquid*

Operating data

☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	*25.0*	*20.0*	*10.0*	°C
p1	*4.08*	*10.0*	*4.08*	bar(g)
○ p2	*3.01*	*2.77*	*2.43*	bar(g)
◉ Δp	*1.07*	*7.23*	*1.65*	bar
◉ qm	*30.0*	*16.0*	*3.0*	kg/h

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ϱ1	☒ *0.0*	☒ *0.0*	☒ *0.0*	kg/m^3
◉ v1	*0.0*		*0.0*	mm^2/s
cF1	*0.0*	*0.0*	*0.0*	m/s

Pipe downstream of valve

Size class downstream of valve	NPS2	*2"*
Schedule downstream	SCH2	*STD*

Valve configuration

Valve type	*Straight globe valve*
Trim type	*Contoured plug*
Flow direction	*FTC*
Valve performance class	*Multi stage valve*
Protection	*Non-hardened*
Low-noise design	*n.a.*

Valve data

Basic characteristic		*Linear*	
☐ Lift stopper			
Selected valve size	NPS	*2"*	
Pressure class	class	*class 600*	
Maximum operating pressure	p,max	*105.0*	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL^2	*0.917*	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	*0.877*	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	*0.604*	-
Valve style modifier (Cv/Cv100 = 1)	Fd	*0.41*	-

Load-dependent values

Noise calculation

Calculation standard (noise, liquid)		*IEC 60534-8-4 (2015)*	

Noise prediction data

Low-noise design		*n.a.*	
Sound velocity	cF1	*0.0*	m/s

Legend

☒ Errors